Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Water-processable Elastic Ionic Polymer Binder for Li-ion Battery			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Battery materials	High polymer materials	Material technology for energy and environmental industry purpose
Concept of Proposed Technology	**Concept and Background** While batteries are key for green energy, most battery research focuses on electrodes and electrolytes, often overlooking binders, especially in high-capacity battery systems. Although binders are tiny components (typically 2–5% of the mass in commercial electrodes), they (a) play multiple critical roles in battery performance and durability; (b) help to disperse the other components in solvent during the fabrication process (with some also acting as a thickener), enabling a homogeneous distribution, (c) hold together the various components of energy storage devices, including the active components, any conductive additive, and the current collector, ensuring all these pieces are kept in contact (Figure 1 shows a schematic for a composite electrode with binder			

interacting with the various components), (d) bridge these separate components through chemical or physical interactions, keeping them together and ensuring mechanical integrity of the electrode without significantly impacting electronic or ionic conductivity, and (e) act as an interface between the electrode and electrolyte, maintaining the stability and performance of battery cells.[1]

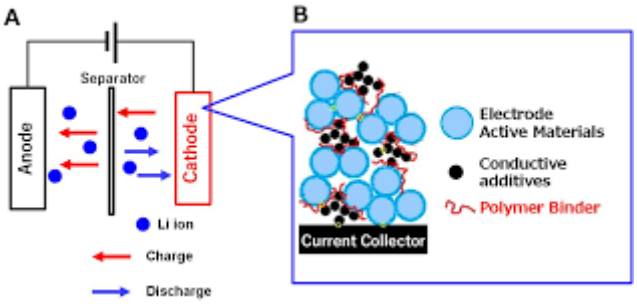

Figure 1. Schematic diagram showing the lithium-ion battery configuration (A) and the conventional electrode composition (B).

The required properties of polymer binders for battery application include (a) good mechanical properties (stiffness, toughness, hardness, and adhesion property to the other components), (b) good thermal stability, (c) excellent chemical electrochemical stability (no reaction with other components during operation), and (d) good dispersive capability for even distribution during fabrication. Furthermore, although not necessary, having electrical and ionic conductivity would greatly improve electrode performance.

Core technical gap (problem statement)

The most common polymer binder for Li ion and other batteries is poly(vinylidene difluoride) (PVDF) because of its good chemical and electrochemical stability and reasonable processability. However, PVDF requires expensive and hazardous organic solvents and does not bind strongly due to its reliance on weak van der Waals forces for interaction with other materials.

Aqueous binders offer several advantages over PVDF in fabricating electrodes including reduced cost and environmental impact. Examples of aqueous polymer binders include styrene-butadiene rubber (SBR), carboxymethylcellulose (CMC), poly(acrylic acid) (PAA), and their derivatives including SBR/CMC blend and polyacrylic latex (LA133). Their chemical structures are shown in Figure 2. Among them, CMC/SBR and LA133 are most extensively used to disperse and bind different materials components of battery. [2]

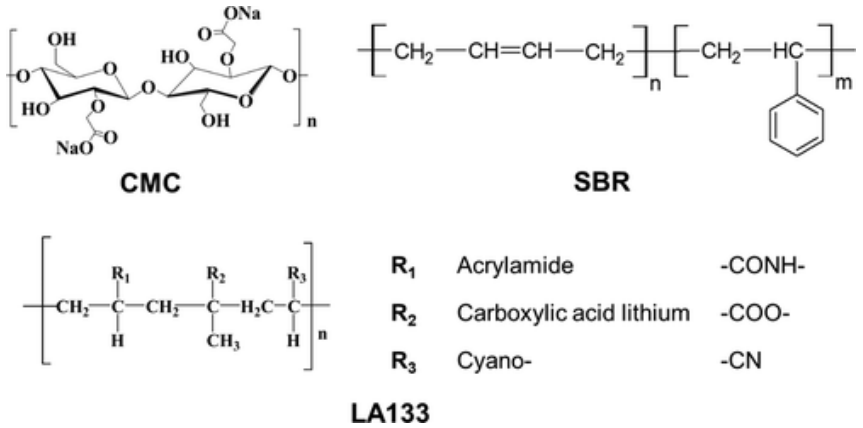

Figure 2. Molecular structures of CMC/SBR and LA122 binders. [2]

Although CMC alone shows improvement over PVDF as a binder, it is typically used by combining with SBR to enhance flexibility and heat resistance. SBR by itself has very low interaction with hydrophobic graphite particles in fabricating anodes, leading to unstable suspensions during preparation. Thus, CMC-SBR is a more practical binder, and it has achieved improved dispersive capabilities, flexibility and binding strength, making them a popular choice for aqueous binder.

RPI Innovation

Polystyrene-*b*-polybutylene-*b*-polystyrene (SBS) is a well-known and widely available thermoplastic triblock copolymer exhibiting microphase separated morphologies at nanoscale due to the incompatibility between the covalently bound polystyrene end blocks and polybutadiene middle block. In this system the higher T_g (~100 °C) polystyrene end block provides mechanical strength to SBS by means of physical crosslinking while the rubbery polybutadiene middle block provides elasticity preventing the material from fracture. The nanophase separated morphology created from the “hard” and “soft” blocks can be advantageous for battery binder applications since such morphology can deliver elastic and tough materials with low tendency to crack under mechanical stress. Furthermore, selective functionalization of one of these blocks can yield high local charge density polymers with ionically conductive channels and mechanically resilient property. Recently, Bae group at RPI has developed sulfonated SBS by incorporating $-SO_3H$ group in the middle block of SBS (Figure 3).

In this proposal, Bae group of RPI proposes development of multifunctional styrene butadiene copolymers that have enhanced interaction with active battery materials by incorporating polar functional groups (-OH and $-SO_3Li$). While the incorporation of $-SO_3$ Li group into polymers would help form hydrogen bonds and enhance ion conductivity, the incorporation of -OH group will further enhance affinity with battery materials. To enhance toughness of copolymers, the RPI team will explore post-functionalization of styrene-butadiene block copolymers (known as SBS) as well as their random copolymers (i.e., SBR). Both SBR and SBS are inexpensive and known to have good elastic mechanical property, which would greatly help accommodate volume changes in charging-

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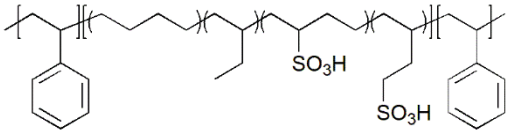
	discharging cycles in battery. The wide commercial availability of SBS (block copolymer) and SBR (random copolymer) in various molecular weights and compositions facilitates the optimization of polymer properties. By selecting and modifying these starting materials appropriately, we can tailor specific performance characteristics through systematic property evaluation.  **H-SBS-SO_3H** Figure 3. Ion conducting styrene-butadiene block copolymer developed by RPI. [3]
Rationale for Joint R&D Project	The RPI team has global reputation for ion-conducting polymer materials development and is looking for a SME collaborator who can provide complementary expertise in water-based binder/nano material slurry formulation engineering and lead the global battery binder market. Despite the current RPI polymer materials have shown good performance and durability in ex-situ conditions (e.g., ion conductivity), their integration into electrodes requires further optimization in design and properties of polymer materials based on feedback from SME's tests. Thus, multiple iterations of material development and feedback are critical for successful development of aqueous binder materials for battery applications. Synergistic effect of this collaboration would result in development of advanced polymer binders that have many attractive features and could lead global market of battery technology.
Key Words	1. Li-ion Battery 2. polymer binder 3. ion conducting polymer

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 5

R&D Objectives

1. RPI: Develop and optimize ion-conducting polymers for ionomer binder of Li ion battery.
2. SME: Evaluate the RPI polymer materials in Li ion battery cathode layer level and suggest outcomes of materials' properties in cell tests
3. RPI and SME: Through multiple iterations of RPI materials test-and-feedback between RPI and SME, optimize materials, electrodes, and other hardware components of battery cell and establish a structure–property map and guidance on scale-up.
4. SME: Demonstrate (a) electrochemical performance and durability at cathode level and cell level and (b) its commercial potential in Li ion battery market.

R&D Target Specifications

	Performance Indicator	Unit	Target	Global Best	Standard/ Certification Criteria
1	Ion conductivity	S/cm	$\geq 10^{-4}$	0.6×10^{-4} (PVDF)	@ 25-40°C
3	Strain	%	≥100	20 (PVDF)	Tensile stress–strain measurement
4	Toughness	GPa	400	320 (PVDF)	Tensile stress x strain value
5	Surface Resistivity	Ω	>1.0E14		35 mm Thickness
6	Resistivity	Ω・cm	>1.0E14		35 mm Thickness

R&D Plan

Task	Period (Month)	Key Tasks & Methodology	Main Deliverables / Milestones
Task 0	0–36	Project governance (joint meeting, reporting), IP/contracting, EHS, quality management (QC)	Kick-off meting; protocol & data management plan; periodic reports; IP strategy; final technical report.
Task 1	0–6	Basic property characterization of current generation RPI polymers materials (battery binder) for electrochemical stability and ionic and electrical conductivity at SME for benchmarking. Optimization of processing conditions (adhesion to the current collector, dispersibility, etc).	Benchmarking report; formulation of Phase 2 plan.
Task 2	3–18	Ion conducting polymer chemistry design: select appropriate combination of backbone and sidechain ionic group for controls of molecular weight, flexibility by tuning hard/soft backbone; try different ionic groups. Characterize relationships of polymer chemical structure and mechanical property, thermal property, and swelling property.	Down-selection of 1st-gen polymer candidates (≥2); structure–property map by linking chemistry variables to membrane properties and SME feedback.
Task 3	7–24	Evaluation of the 1st-gen polymer candidates: adhesion to the current collector and cohesion between active materials and conductive agents, mechanical flexibility and elasticity, electrochemical & chemical stability, dispersity in water-based slurry, and ionic and electrical conductivity.	Interim benchmarking report

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	Task 4	19–30	Optimizations of polymer synthetic process and casting of down-selected 1st-gen candidates: molecular weight control, solvents selection, isolation process, casting parameter, thickness optimization.	Down-select 1 polymer material candidate.
	Task 5	25–36	Optimization of binder property on the electrode: mechanical properties and electrochemical stability, dispersibility, and electrical conductivity in battery cell	Final benchmarking report, Scale-up and production plan
	Task 6	30–36	Scale up and commercialization planning: cost model, supply chain, production scale-up roadmap, and customer engagement for adoption.	Go-to-market plan; pilot production plan; partner/customer LOI (if available).

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Polymer material design, synthesis, and characterization: Optimize polymer architecture and synthetic chemistry to balance ion conductivity, mechanical property, and processability.
	• Advanced characterization of polymers by NMR, FT-IR, viscoelasticity (DMA/rheology), morphology, thermal property (DSC/SEM as needed).
Expected R&R of Korean SME Partner	• Electrode engineering based on RPI water-based anode binder polymer. • Development of electrodes with high colloidal stability for uniform coating on the current collector. • Development of stable dispersion formulation ability to uniformly disperse active materials and conductive agents in water-based slurry to prevent agglomeration with binder materials. • Characterization of electrodes and Li ion battery performance and long-term durability.
	• Identify end-use customers and expand high-performance Li+ battery market globally. • Lead commercialization: product spec, cost/production plan, and regulatory/safety documentation.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Outcomes:** • Development of water-processable polymer binder materials that could serve as alternative to PVDF in global battery market. • Optimization of multiple materials properties, manufacturing processes, and customization of binder materials for battery technologies and customers. • Mapping of cost-process relationships of binder materials and establishment of guideline of cost reduction. **Impact:** • Establish a secure global supply chain for key materials of battery binder that have potential of use in global market production through joint IP development. • Meet the growing demand of global battery market with lower cost, more environmentally friendly material products which can accelerate the market. • Boost the global visibility and market share of Korean SMEs in battery sector by creating robust international value chains.

6. References

[1] B. Chen, Z. Zhang, M. Xiao, S. Wang, S. Huang, D. Han, Y. Meng, *ChemElectroChem*, **2024**, *11*, e202300651.
[2] W. Wang, X. Yue, J. Meng, X. Wang, Y. Zhou, q. Wang, Z. Fu, *J. Phys. Chem. C.* **2019**, *123*, 250–257.
[3] M. K. Pagels, D. Tian, S. Turan, C. Bae, *Polymer Chem.* **2025**, *16*, 3977–3985.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization